Why $D_\ell^{\rm rec}$ is not D_ℓ^{yy} plus a weight-filtered residual auto-spectrum

Abstract

Harmonic ILC is linear in the maps. The angular power spectrum is not. Therefore

$$D_\ell(\hat{y}) \;\neq\; D_\ell(y) \;+\; D_\ell\big(F[\mathrm{CIB}+\mathrm{CMB}+n]\big)$$

in general. The identity that *does* hold is the quadratic expansion of $D_\ell(\hat{y})$ in the filtered components of the same operator F applied to the same inputs. The plotted quantity $(D_\ell^{\rm rec} - D_\ell^{yy})/D_\ell^{yy,\rm fid}$ is that difference for the HILC map against the noiseless Compton-y map. It is not a bug; it is the remainder after identifying "signal" with y rather than with $F[\mathrm{tSZ}]$.

1 The HILC operator

Write the six frequency maps as

$$d_\nu \;=\; B_\nu\big(s^{\rm CMB} + s_\nu^{\rm tSZ} + s_\nu^{\rm CIB}\big) \;+\; n_\nu, \tag{1}$$

with B_ν the channel Gaussian beam and n_ν homogeneous white noise (unbeamed). For a given sky (full sky or $q > 5$) and deprojection, pyILC's top-hat harmonic ILC is a diagonal multiplier in harmonic space. On the full sky,

$$a_{\ell m}(\hat{y}) \;=\; \sum_\nu F_\nu(\ell)\, a_{\ell m}(d_\nu), \qquad F_\nu(\ell) \;=\; w_\nu(\ell)\, T(\ell)\, \frac{B_{10}(\ell)}{B_\nu(\ell)}, \tag{2}$$

where $w_\nu(\ell)$ is piecewise constant on bins of width $\Delta\ell = 50$, $T(\ell)$ is the edge taper, and B_{10} is the $10'$ common beam. On the $q > 5$ sky the same form holds after replacing d_ν by Md_ν and w_ν by the weights estimated from those masked maps:

$$a_{\ell m}(\hat{y}^{q5}) \;=\; \sum_\nu F_\nu^{q5}(\ell)\, a_{\ell m}(Md_\nu). \tag{3}$$

Pixel multiplication by M and the harmonic filter F^{q5} do not commute. That is intended: the $q > 5$ reconstruction is a re-run of ILC on the cut sky, not the full-sky $\hat{y}$ multiplied by M.

In either case the reconstruction is linear,

$$\hat{y} \;=\; F[d] \;=\; F[\mathrm{tSZ}] + F[\mathrm{CIB}] + F[\mathrm{CMB}] + F[n]. \tag{4}$$

Call $t \equiv F[\mathrm{tSZ}]$ and $r \equiv F[\mathrm{CIB}] + F[\mathrm{CMB}] + F[n]$, so $\hat{y} = t + r$.

2 Power spectra are quadratic

Let $D_\ell(\,\cdot\,)$ denote the binned $\ell(\ell+1)C_\ell/2\pi$ obtained from a noise-debiased split-cross ($r_1 \times r_2$) after MASTER decoupling and deconvolution of $B_{10}^2 w_{\rm pix}^2$. Then

$$D_\ell(\hat{y}) \;=\; D_\ell(t+r) \;=\; D_\ell(t) + D_\ell(r) + 2\, D_\ell(t,r). \tag{5}$$

The cross $D_\ell(t,r)$ expands into every mixed pair among tSZ, CIB, CMB and noise, each counted once as $C_\ell(a_{r_1}, b_{r_2}) + C_\ell(b_{r_1}, a_{r_2})$. There is no extra factor of two beyond that.

Proposition 1. *If t and r are obtained by applying the* same *F to a partition of the* same *inputs that produced $\hat{y}$, then* (5) *is an identity: the autos of the four components plus those six crosses equal $D_\ell(\hat{y})$ to machine precision.*

That is the only sense in which "total = signal + residual". The residual that belongs in the identity is $D_\ell(r) + 2D_\ell(t,r)$, not $D_\ell(r)$ alone, and the signal is $D_\ell(t)$, not $D_\ell(y)$.

3 The noiseless Compton-y map is not t

The plotted black curves use the MASTER auto-spectrum of the input Compton-y map (pixel-window deconvolved). Call it D_ℓ^{yy}. The ILC constraint is

$$\sum_\nu w_\nu(\ell)\, a_\nu^{\rm tSZ} \;=\; 1 \tag{6}$$

on the tSZ mixing vector, so t is a $10'$-beamed, tapered copy of y, not y itself. After deconvolving $B_{10}^2 w_{\rm pix}^2$ one has $D_\ell(t) \approx D_\ell^{yy}$ on the full sky (measured ratio 1.003 at $\ell = 30.5$ for $M_* - 1\sigma$, CIB+$\delta\beta$). The equality is not algebraic:

$$D_\ell(\hat{y}) - D_\ell^{yy} \;=\; \underbrace{\left[D_\ell(t) - D_\ell^{yy}\right]}_{\text{tSZ mismatch}} + D_\ell(r) + 2\, D_\ell(t,r). \tag{7}$$

The left-hand side is the plotted residual. It equals a weight-filtered foreground auto-spectrum only if $D_\ell(t) = D_\ell^{yy}$ *and* the crosses $2D_\ell(t,r)$ are included in that residual.

On the $q > 5$ sky an extra mismatch appears: D_ℓ^{yy} is the full y map masked only inside MASTER, whereas $t^{q5} = F^{q5}[M\, s^{\rm tSZ}]$. Because $F^{q5}M \neq MF^{\rm full}$, $D_\ell(t^{q5})$ need not track the masked D_ℓ^{yy} even if (6) holds. Empirically, at $\ell = 30.5$ for $M_* - 1\sigma$, CIB+$\delta\beta$,

$$\begin{aligned} D_\ell(\hat{y}^{\rm full}, M \text{ at spectrum only}) &= 6.37 \times 10^{-15}, \\ D_\ell^{yy}(M \text{ at spectrum only}) &= 5.74 \times 10^{-15}, \\ D_\ell(\hat{y}^{q5}, M \text{ at spectrum}) &= 1.59 \times 10^{-14}. \end{aligned}$$

The first two agree at the ten-percent level. The third is the re-run, and is larger because F^{q5} is a different operator. That difference is physical for the $q > 5$ HILC, not an arithmetic error.

4 A different operator does not close the identity

A harmonic reconstruction $y_{\rm coadd}$ obtained by reading the saved w_ν, applying (2) to component alms, and summing, obeys Proposition 1 for *that* operator: $D_\ell(y_{\rm coadd}) = D_\ell(t_{\rm coadd} + r_{\rm coadd})$. If the component alms were masked differently from (3) (mask applied twice, or applied after filtering), then $F_{\rm coadd} \neq F^{q5}$ and

$$D_\ell(\hat{y}) - D_\ell(y_{\rm coadd}) \;=\; D_\ell(\hat{y} - y_{\rm coadd}) + 2\, D_\ell(\hat{y} - y_{\rm coadd},\, y_{\rm coadd}). \tag{8}$$

This gap is not missing power in (5). It is the spectrum of two different fields. On the full sky $F_{\rm coadd} \approx F$ and the gap collapses ($\sim 0.02\, D_\ell^{yy}$ at $\ell = 30.5$). On $q > 5$ it does not, because the coadd never implemented (3).

5 What the figure shows

For each astrophysical variant and each sky,

$$\text{black:} \quad \frac{D_\ell^{yy,\mathrm{var}}}{D_\ell^{yy,\mathrm{fid}}} - 1, \tag{9}$$

$$\text{colour:} \quad \frac{D_\ell^{\mathrm{rec,var}} - D_\ell^{yy,\mathrm{var}}}{D_\ell^{yy,\mathrm{fid}}}. \tag{10}$$

Adding them recovers $(D_\ell^{\mathrm{rec,var}}/D_\ell^{yy,\mathrm{fid}}) - 1$, which is the fractional change of the HILC map relative to the fiducial noiseless y. It is *not* required to equal the fractional weight-filtered CIB auto-spectrum.

The shaded bands are $\pm\sigma/D_\ell^{yy,\mathrm{fid}}$ with $\sigma^2 = \sigma_G^2 + \sigma_T^2$ from the fiducial HILC split-cross of that same deprojection and sky. They are uncertainties on the reconstruction, normalised to the same denominator as the curves.

Remark 1. *The statement "reconstructed power = signal + residual" is true if and only if (i) signal means $D_\ell(F[\mathrm{tSZ}])$, (ii) residual means $D_\ell(r) + 2D_\ell(t, r)$, and (iii) F and the inputs are those that produced $\hat{y}$. Replacing (i) by D_ℓ^{yy} or (iii) by a different filter makes the equality false without contradicting linearity.*